

ALP for 8085uP

Prob 1: WAALP for 8085uP which will unpack the data available in D000h memory location and put the result in D001h (HON) and D002h (LON) location

Soln

1	LDA D000H
2	MOV B, A
3	ANI 0FH
4	STA D002H
5	MOV A, B
6	ANI FOH
7	RLC
	RLC
	RLC
9	RLC
10	STA D001H
11	RST1

D000H: B5H

D001H: 0BH

D002H: 05H

(1) $(A) \leftarrow D000H$

(2) $(B) \leftarrow (A) = \underline{B5H}$

(3) $(A) \leftarrow 05H$

(4) $D002H \leftarrow (A)$
05H

(5) $(A) \leftarrow B = B5H$

(6) $(A) \leftarrow B0H$

(7)-(9) $(A) \leftarrow 0BH$

(10) $D001H \leftarrow (A)$

1	LXI H, D002H
2	MOV B, M
3	DCX H
4	MOV A, M
6	RLC
7	RLC
8	RLC
9	RLC
10	ORA B
11	DCX H (12) MOV M, A
	MOV M, A
	RST1
PACKING THE SAME DATA	

Packing

D002H - 0AH

D001H - 02H

D000H - 2AH

(1) $(HL) \leftarrow D002H$
(2) $(HL) \leftarrow * (HL)$
= 0AH

(3) $(HL) \leftarrow D001H$

(4) $A \leftarrow * (HL)$
= 02H

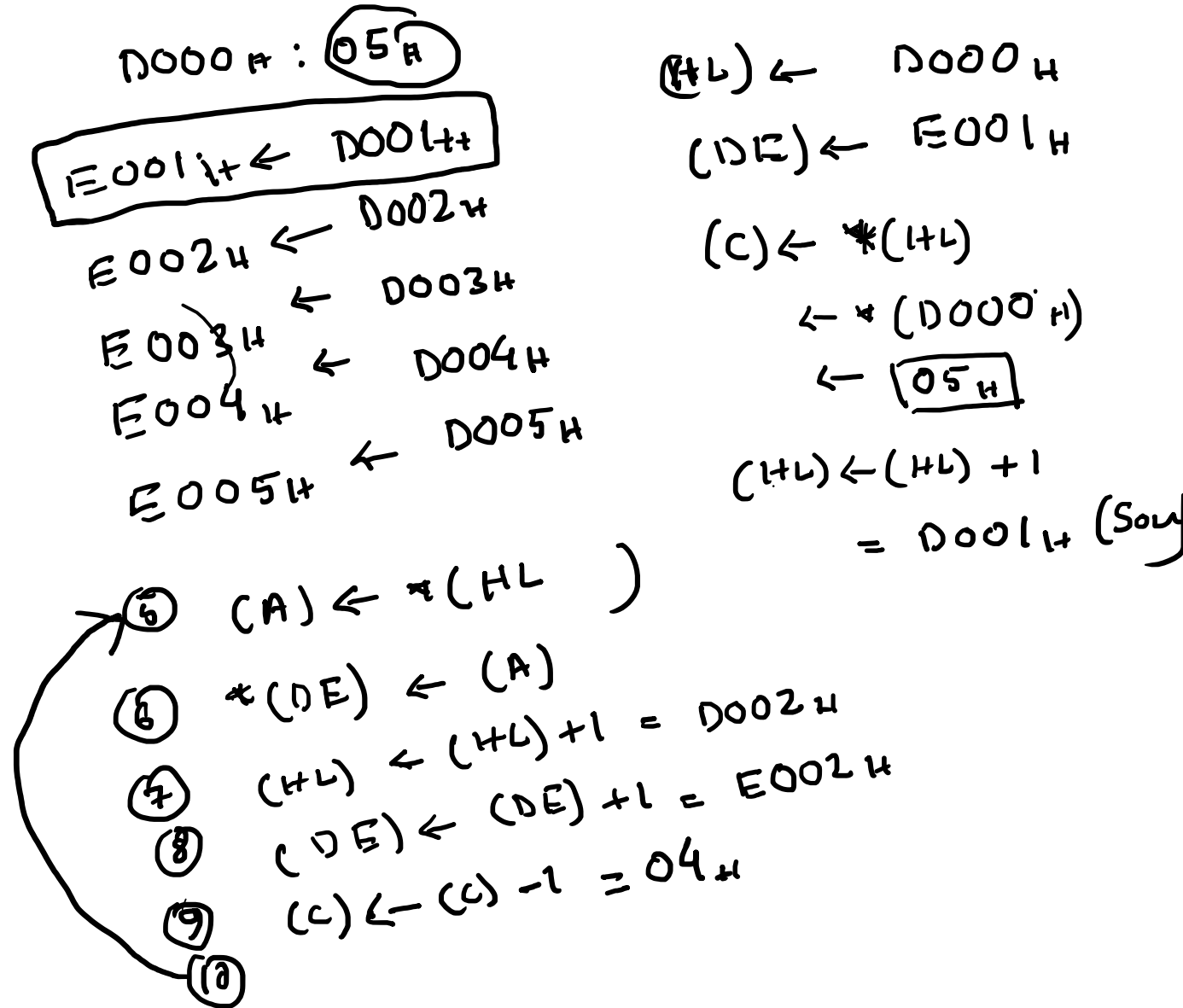
(6)-(9) $A \leftarrow 20H$

(10) $A \leftarrow A \text{ OR } B$
20H OR 0AH

(11) $(HL) \leftarrow D000H$
(12) $* (HL) \leftarrow A = 2AH$

Prob 2: WAALP for 8085uP which will COPY THE CONTENT SERIES OF DATA STARTING FROM D001H TO E001H WHERE THE NO OF DATA AVAILABLE IN D000H

	LXI H, D000H	
	LXI D, E001H	
	MOV C, M	
②	INX H	
L1:	MOVA, M	⑤
	STAX D	⑥
	INX H	⑦
	INX D	⑧
	DCRC	⑨
	JNZ L1	10
	RST1	



Prob 3: WAALP for 8085uP to take a Hexadecimal number stored in D001h location and convert it to its equivalent BCD value. Store the result in D002h location. Consider the number in D001h is not more than ~~16~~ 15 in decimal.

①	LDA D001H
②	CPI <u>0AH</u>
③	JC L1
	ADI 06H ④
L1:	STA D002H ⑤
	RST1

$$17_{10} = 0001\ 0001_2 = 11$$

JNC

$$0_{10} - 9_{10} = 00H - 09H$$

$$10_{10} - 15_{10} = 0AH - 0FH$$

$$D001H = \boxed{07H}$$

D001H
11
0BH

① $(A) \leftarrow * (D001H) = 07H \mid 0BH$

② A will be compared with 0AH (10₁₀)

$$A \geq 0AH = CY0 \quad Z=0$$

$$A = 0AH = CY0 \quad Z=1$$

$$A < 0AH = CY1 \quad Z=0$$

④ $(A) \leftarrow A + 06H$
 $\leftarrow 0BH + 06H$
 $(A) \leftarrow 11$

⑥ $* (D002H) \leftarrow (A) = 07H$

Prob 4: WAALP for 8085uP to MULTIPLY TWO 8-BITS NUMBERS PRESENT IN C001H AND C002H LOCATIONS. STORE THE RESULT IN D001H (HOB) AND D002H (LOB) LOCATIONS, RESPECTIVELY.

	XRA A
	MVI D, 00H
	LXI H, C001H
	MOV C, M
	INX H
	MOV B, M
L2:	ADD B
	JNC L1
	INR D
L1:	DCR C
	JNZ L2
	STA D002H
	MOV A, D
	STA D001H
	RST1

Prob 5: WAALP for 8085uP to DIVIDE A 8-BITS NUMBER IN C001H BY ANOTHER 8-BITS NUMBER IN C002H LOCATIONS. STORE THE RESULT IN D001H (QUOTIENT) AND D002H (REMINDER) LOCATIONS, RESPECTIVELY.

	XRA A
	MVI D, 00H
	LXI H, C001H
	MOV C, M
	INX H
	MOV B, M
L2:	SUB B
	JC L1
	INR D
	JMP L2
L1:	ADD B
	STA D002H
	MOV A, D
	STA D001H
	RST1

Prob 6: CALCULATE THE TIME TAKEN FOR THE CODE TO COMPLETE IF THE 8085uP IS WORKING IN I. 3MHz AND II. 5MHz CLOCK FREQUENCY

LABEL	INSTRUCTION	T-STATE
	MVI D, 04H	7
L2:	LXI B, FFAAH	10
L1:	DCX B	6
	NOP	4
	NOP	4
	NOP	4
	NOP	4
	MOV A, C	4
	ORA B	4
	JNZ L1	10
	DCR D	4
	JNZ L2	10
	RST1	4

Total T-State:

$$7 + 4 \times (10 + 65450 \times (6 + 4 + 4 + 4 + 4 + 4 + 4 + 10)) + 4$$

Prob 6: WAALP TO GENERATE A DELAY OF 0.25SEC FOR 8085uP WORKING IN 5MHz CLOCK FREQUENCY.

LABEL	INSTRUCTION	T-STATE
	LXI B, XXXXH	10
L1:	DCX B	6
	NOP	4
	NOP	4
	NOP	4
	NOP	4
	MOV A, C	4
	ORA B	4
	JNZ L1	10
	RST1	4

Total T-State:

$$10 + x * (6 + 4 + 4 + 4 + 4 + 4 + 4 + 10) + 4$$